NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

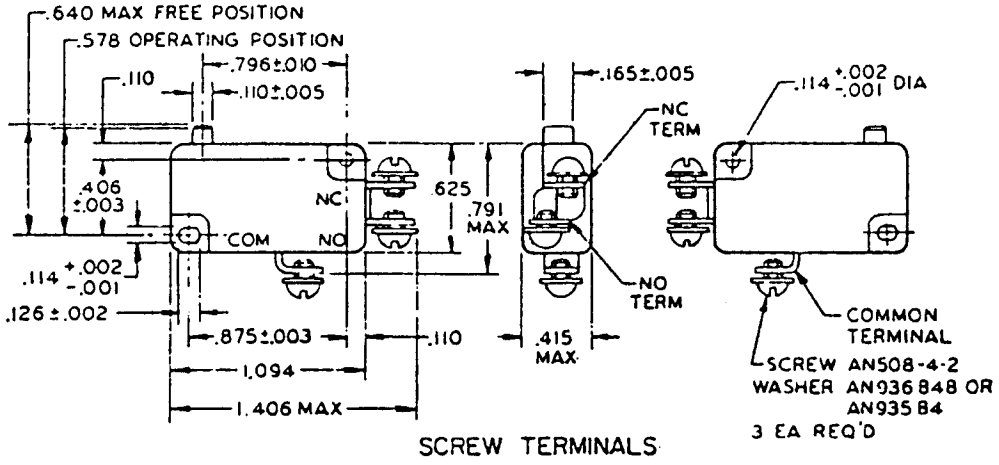
MIL-PRF-8805/27C
5 August 1985
SUPERSEDING
MIL-S-8805/27B
11 July 1972

PERFORMANCE SPECIFICATION SHEET

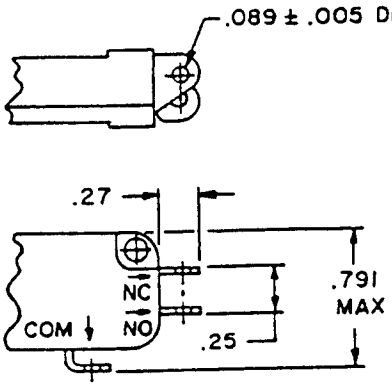
SWITCHES, SENSITIVE, 10 AMPERES
AND LOW LEVEL, UNSEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.



SCREW TERMINALS



SOLDER TERMINALS

INCHES	MM	INCHES	MM
.001	0.03	.270	6.86
.002	0.05	.406	10.31
.003	0.08	.415	10.54
.005	0.13	.578	14.68
.010	0.25	.625	15.88
.089	2.26	.640	16.26
.110	2.79	.791	20.09
.114	2.90	.796	20.22
.126	3.20	.875	22.23
.165	4.19	1.094	27.79
.250	6.35	1.406	35.71

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.015 (0.38 mm).
3. Exact shape of switch optional provided dimensions specified are not exceeded.
4. Mark, stamp, or mold NO, NC and COM terminal designations and arrows on either side of switch.
5. Metric equivalents are given for general information only.

FIGURE 1. Configuration and dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Enclosure design. 1 (unsealed).

Temperature characteristic. 1 (-55°C to +85°C).

Shock type: Method 213, test condition A, MIL-STD-202 (50 G).

Vibration grade. 1 (10 to 500 Hz).

Weight .30 ounce maximum.

Operating characteristics

Actuating force·

MS25253-1, -2, -3, and -5 through -7	14 ounces maximum.
MS25253-8 through -13	135 grams maximum.

Movement differential.

MS25253-1, -2, -3, and -5 through -7·	.020 inch maximum.
MS25253-8 through -13	.010 inch maximum.

Pretravel .047 inch maximum.

Overtravel.

MS25253-1, -2, -3, and -5 through -7	.031 inch minimum.
MS25253-8 through -13	.050 inch minimum.

Strength of actuating means 25 pounds.

Material

Plunger May be glass filled nylon.

Contact resistance

MS25253-1, -2, -3, and -5 through -7	Not applicable.
MS25253-8 through -13	0.05 ohm maximum initially.
	0.10 ohm maximum after mechanical life.

Contact bounce

MS25253-1, -2, -3, and -5 through -7	Not applicable.
MS25253-8 through -13	5 milliseconds maximum.

Dielectric withstanding voltage

Sea level	1,000 V rms.
Altitudes	400 V rms at 50,000 ft. In qualification inspection table, after electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.

Mechanical endurance.

MS25253-1, -2, -3, and -5 through 7·	25,000 cycles.
MS25253-8 through -13	500,000 cycles.

Electrical endurance (power circuits) 25,000 cycles.

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Low level circuit:

MS25253-1, -2, -3, and -5 through -7: Not applicable.
MS25253-8 through -13: 50,000 cycles.

Electronic logic circuit (5.0 V dc, .010 ampere):

MS25253-1, -2, -3, and -5 through -7: Not applicable.
MS25253-8 through -13: 50,000 cycles when tested in accordance with ANSI/EIA RS-448, method 17. Rate of action shall be 120 cycles per minute maximum. No "stick" or "misses" allowed.

Operating temperature.

a. 25 percent of the test cycle at the minimum temperature specified.
b. 25 percent of the test cycle at room ambient conditions.
c. 50 percent of the test cycle at the maximum temperature specified.

Electrical ratings. See table I and II.

Quality assurance:

Qualification inspection. See table III.

Inspection of product for delivery. See table IV.

Part numbers: See table V.

TABLE I. Electrical ratings silver contacts (MS25253-1, -2, -3, and -5 through -7).

Load	Sea level		50,000 feet	
	28 V dc	115 V ac, 60 Hz	28 V dc	115 V ac, 60 Hz
	Amperes	Amperes	Amperes	Amperes
Resistive	10	10	10	10
Inductive	10	10	6	10
Motor	6	3	---	---

TABLE II. Electrical ratings gold contacts (MS25253-8 through -13).

Load	Sea level and 50,000 feet		Sea level
	Power circuit 28 V dc	Electronic logic circuit 5 V dc	Low level 1/ 30 millivolts
	Amperes	Amperes	Amperes
Resistive	1.0	0.01	0.01 max
Inductive	0.5	---	---

1/ 30 millivolts max dc or peak ac.

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TABLE III. Qualification inspection (group submission).

Inspection	Sample	Extent of approval
Group I Visual and mechanical 1/ Solderability (when applicable) 2/ Dielectric withstanding voltage Insulation resistance Operating characteristics	 MS25253-1 (all samples) MS25253-2, -3 (2 samples) each visual and mechanical MS25253-11 (all samples)	
Group II Terminal strength 3/ Strength of actuating means 3/ Thermal shock Vibration Shock Moisture resistance Marking visibility Dielectric withstanding voltage Operating characteristics	 MS25253-1, -11 (4 samples each)	
Group IV Salt spray Marking visibility	 MS25253-1 (2 samples)	
Group V Explosion Operating characteristics	 MS25253-1 (2 samples)	
Group VI Contact resistance (when applicable) Contact bounce (when specified) Low temperature operation Mechanical endurance at low temperature Mechanical endurance at high temperature Contact resistance (when applicable) Short circuit 4/ Dielectric withstanding voltage Operating characteristics	 MS25253-1, -11 (4 samples each)	all
Group VII Overload cycling Electrical endurance Contact resistance (when applicable) Dielectric withstanding voltage Operating characteristics	 MS25253-1 (20 samples) MS25253-11 (8 samples)	
Group X Low level circuit Operating characteristics	 MS25253-11 (2 amperes)	
Electronic logic circuit Operating characteristics	 MS25253-11 (2 samples)	

1/ Two samples for physical dimensions.

2/ Four samples of each applicable terminal type.

3/ Two samples of each part number.

4/ Short circuit test shall be conducted at the 28 V dc electrical rating.

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TABLE IV. Inspection of product for delivery.

Inspection	AQL (percent defective) Major	Minor
Visual and mechanical 1/	1.0	4.0
Operating characteristics 3/	1.0	---
Contact bounce (when specified) 3/	--- 2/	---
Dielectric withstanding voltage	0.65	---
Contact resistance 3/	0.65	---

1/ Inprocess inspection may be used to satisfy the dimensional requirements.
2/ When applicable, two samples per lot.
3/ Do not exceed 6 V dc, or peak ac, 10 milliamperes during inspection of MS25253-8 through -13.

TABLE V. Part numbers.

Part number 1/ MS25253-	Description					Suggested applications 2/		
	Contacts	Circuit	Rating maximum	Terminals Type	Terminals Strength (lbs)	Power Circuit	Electronic logic	Low level
1	Silver	SPDT	Table I (10 AMPS)	Screw	20	X		
2	"	SPNO	"	"	"	X		
3	"	SPNC	"	"	"	X		
5	"	SPDT	"	Solder	"	X		
6	"	SPNO	"	"	"	X		
7	"	SPNC	"	"	"	X		
8 3/	Gold	SPDT	Table II (1 AMP)	Screw	"	X	X	X
9 "	"	SPNO	"	"	"	X	X	X
10 "	"	SPNC	"	"	"	X	X	X
11 "	"	SPDT	"	Solder	"	X	X	X
12 "	"	SPNO	"	"	"	X	X	X
13 "	"	SPNC	"	"	"	X	X	X

1/ MS25253-4 is canceled, use MS25253-1.
2/ Application information

Power circuits - Those electrical loads where the voltage and current exceed the minimum arcing conditions of the contact material. As a general rule, application loads in excess of 8 volts, 0.5 amperes are considered power circuits.

Electronic Logic Circuits - Those electrical loads in which the applied voltage is less than the arcing voltage and greater than the non-arcing application loads in excess of 0.5 volts are considered logic level circuits.

Low Level Circuits - Those electrical loads in which the applied voltage is less than the softening voltage of the contact material. As a general rule, application loads less than .08 volts are considered low level circuits.

3/ MS25253-8 through -13 have two levels of capabilities. However, switches previously tested or used above 10 milliamperes resistive at 6 V dc max or peak ac are not recommended for use in electronic logic or low level application.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS, SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AV, MI
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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